Stage 1 – Current Camera Preview Flow

Technical Assessment · Snapshot of the existing implementation (no blur pipeline yet)

Overview

The app is a Next.js 15 / React 19 client-rendered page. `app/page.tsx` renders a header ("Realtime Background Blur") and a single `CameraPreview` component. There is no backend and no video processing yet; the component displays the raw front-camera feed.

Components

File	Role
`app/page.tsx`	Page shell and header; renders `CameraPreview`
`components/CameraPreview.tsx`	Camera lifecycle, state machine, error messages, UI
`components/ui/button.tsx`	Shared button (`glass` and `outline` variants)

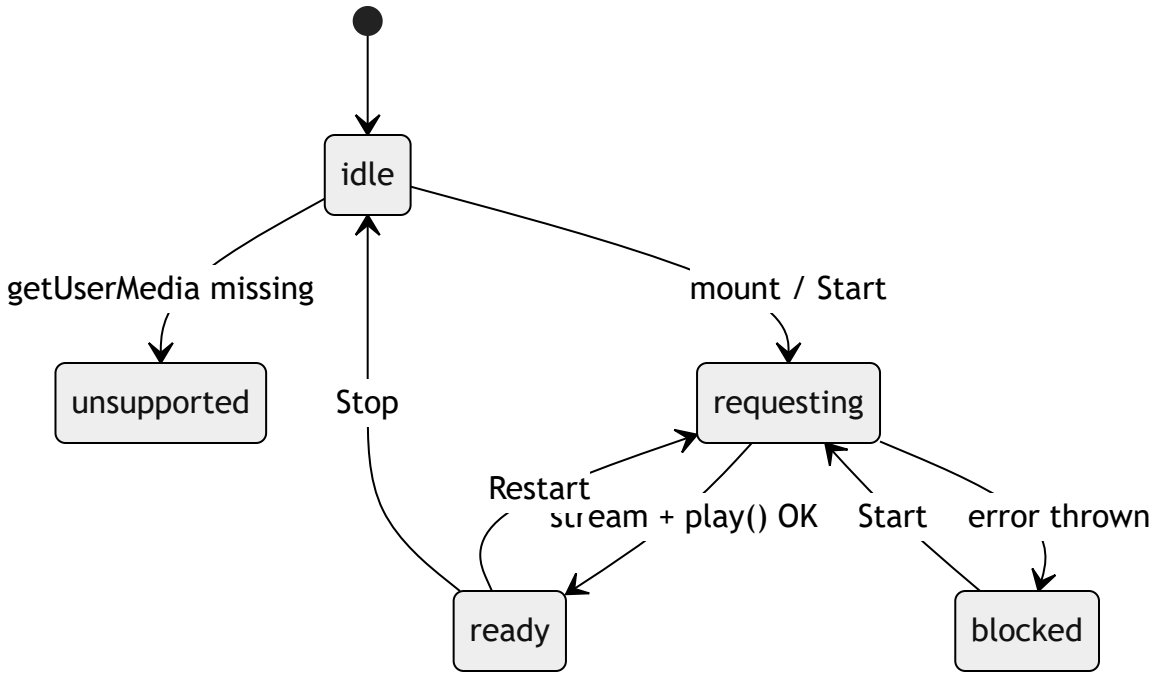
Camera Configuration

- `audio: false`
- `video.facingMode: "user"` (front camera)
- `video.width: { ideal: 1280 }`, `video.height: { ideal: 720 }`

States

State	Meaning	Message shown
`idle`	Not started or stopped	"Waiting for camera permission." / "Camera stopped."
`requesting`	Waiting on the browser permission prompt	"Requesting front camera."
`ready`	Stream attached and playing	"Front camera is live."
`blocked`	`getUserMedia` or `play()` threw	Error-specific message (see Error Handling)
`unsupported`	`navigator.mediaDevices.getUserMedia` is missing	"This browser does not support camera capture."

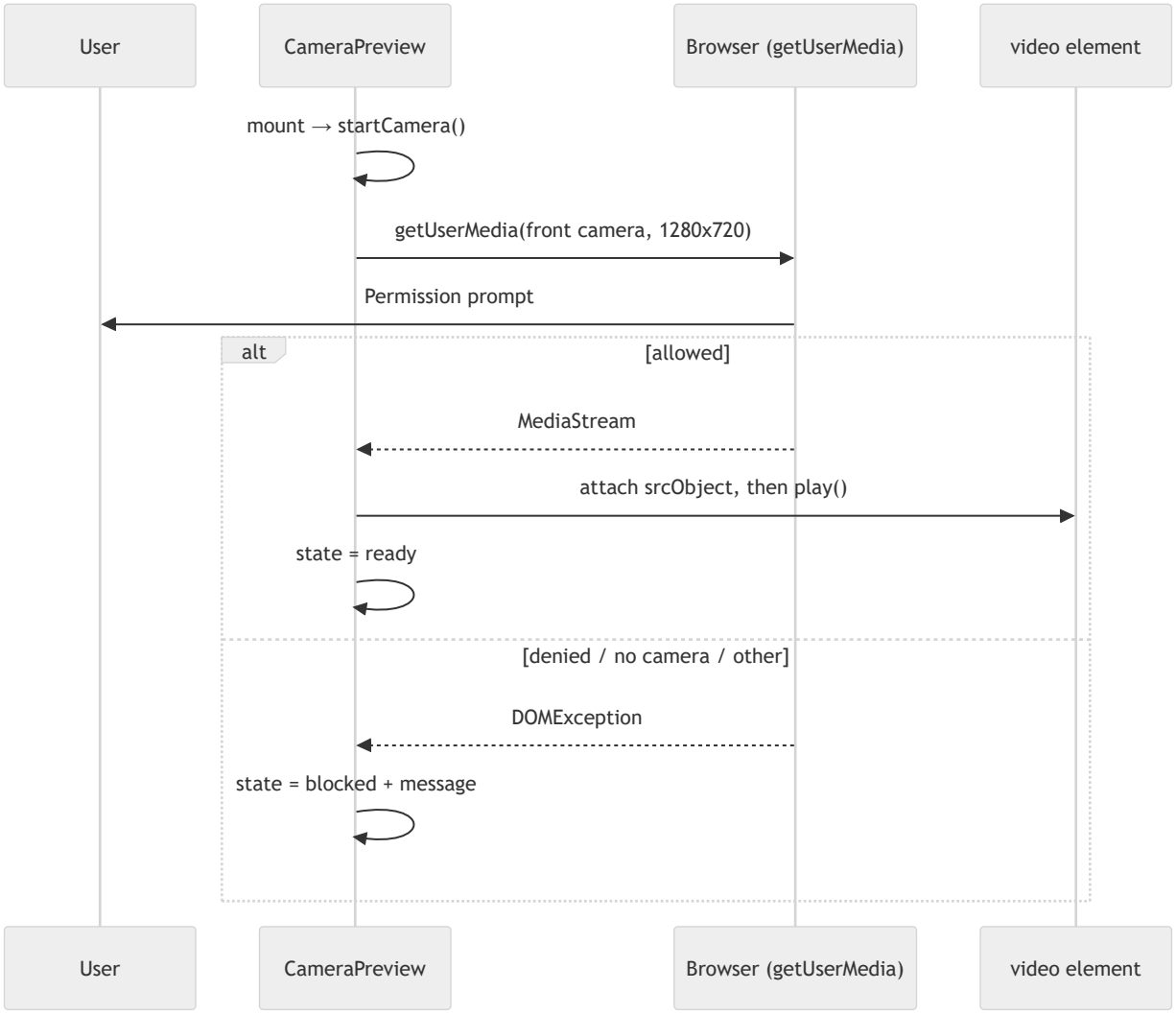
State diagram



Flow

1. Component mounts → `useEffect` calls `startCamera()` automatically.
2. `startCamera()`:
   a. If `navigator.mediaDevices?.getUserMedia` is missing → state `unsupported`, return.
   b. Set state `requesting`.
   c. Stop any existing stream and clear `video.srcObject`.
   d. Call `getUserMedia(cameraConstraints)`.
   e. On success: store the stream in `streamRef`, attach it to `<video>`, `await video.play()`, set state `ready`.
   f. On failure: stop and clear the stream, set state `blocked`, show `getErrorMessage(error)`.
3. **Start / Restart** button → calls `startCamera()` again (disabled while `requesting`).
4. **Stop** button → `stopCamera()` stops all tracks, clears `srcObject`, sets state `idle` (disabled unless `ready`).
5. Unmount → effect cleanup stops all tracks.

Sequence: startup



Error Handling (Current)

Error	Handled?	Message
`getUserMedia` API missing	Yes	"This browser does not support camera capture."
`NotAllowedError`	Yes	"Camera permission was blocked. Allow camera access in the browser to continue."
`NotFoundError`	Yes	"No camera was found on this device."
`NotReadableError`, `OverconstrainedError`, `SecurityError`, `AbortError`, `play()` rejection	Generic	"The camera could not be started."
Track `ended` mid-session (camera unplugged / permission revoked)	No	UI stays in `ready`

UI

- Status area uses `aria-live="polite"`; error text is styled with `camera-error`.
- `<video>` is `muted`, `playsInline`, `autoPlay`.
- A placeholder overlay is shown whenever state ≠ `ready` ("Waiting for the browser permission prompt." while requesting, otherwise the current message).

Not Yet Implemented

- Background segmentation and blur (the stated goal of Stage 1).